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The combination of aerobic and microaerobic promote hydrolysis and acidification of rice straw and pig manure: Balance of insoluble and soluble substrate

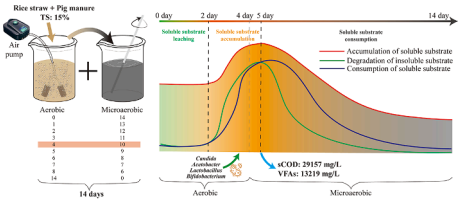
Shuyu Xiang [a], Yuhuan Liu [a], Feihu Lu [a], Qi Zhang [a], Yunpu Wang [a], Jianghua Xiong [b], Zhenxia Huang [b], Zhigang Yu [c], Roger Ruan [d], Xian Cui [a,*]

[a] State Key Laboratory of Food Science and Technology, Engineering Research Center for Biomass Conversion, Ministry of Education, Nanchang University, Nanchang, Jiangxi 330047, PR China
[b] Agricultural Ecology and Resources Protection Station of Jiangxi Province, Nanchang 330046, China
[c] Advanced Water Management Centre, The University of Queensland, Brisbane 4072, Australia
[d] Center for Biorefining and Department of Bioproducts and Biosystems Engineering, University of Minnesota, Paul 55108, USA

HIGHLIGHTS

- Balance of ISS degradation and SS consumption under high TS during HA was studied.
- HA consisted of three period: SS dissolution, ISS degradation and SS consumption.
- Balance point in D4 group was 5th day with 29157 mg/L sCOD and 13219 mg/L VFAs.
- *Candida, Lactobacillus, Bifidobacterium,* and *Acetobacter* led to SS accumulation.
- Combination of aerobic and microaerobic can effectively pretreat lignocellulose.

GRAPHICAL ABSTRACT



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ABSTRACT

Separated hydrolysis and acidification is an effective pretreatment method for anaerobic digestion of lignocellulose. However, excess consumption of soluble substrates remains a problem. Rice straw and pig manure were used as substrates with biogas slurry as the inoculum, combined with aerobic and microaerobic conditions in the 14-day hydrolysis and acidification. Aeration can significantly accelerate volatile solid degradation (38.25%), especially the lignocellulose. Soluble chemical oxygen demand (29157 mg/L) and volatile fatty acids (13219 mg/L) of the group with 4 days aerobic treatment, reached their peaks on day 5, obtaining a balanced insoluble substrate degradation and soluble substrate consumption. *Candida, Lactobacillus, Bifidobacterium,* and *Acetobacter* were enriched at the balanced point for positive contribution to the degradation of the insoluble substrate and the generation of soluble substrate. This study not only reveals the balance between degradation and consumption, but also provides new insight into biogas slurry recycling and anaerobic digestion precursor substrate production.

* Corresponding author at: 235 Nanjing Dong Road, Nanchang 330047, China.
E-mail address: cuixian@ncu.edu.cn (X. Cui).

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1. Introduction

Every year, China generates approximately 48 million tons of rice straw (RS) and 240 million tons of pig manure (PM) (Shaheen et al., 2022; Xiao et al., 2022), and its resource efficiency remains insufficient. There are various resource recovery technologies available to reduce the amount of solid waste to avoid a large amount of CO_2 emissions caused by direct combustion, such as anaerobic digestion (AD), pyrolysis, composting, and others. AD is a better treatment option for these biomass resources because of its multiple properties, such as energy, fertilizer, and positive contribution to "carbon neutral" (Gross et al., 2021; Rodin et al., 2020). Anaerobic co-digestion (co-AD) of RS and PM to produce biogas is currently a popular treatment method because it can compensate for the imbalance in the carbon to nitrogen ratio compared to a single substrate, as well as reduce ammonia inhibition in PM and increase microbial abundance (Xu et al., 2018). However, co-AD has yet to overcome the time constraint (Qu et al., 2021; Wang et al., 2020; Zhang et al., 2019), and it takes longer compared to other techniques of waste reduction.

The hydrolysis stage is the time-limiting stage in AD of lignocellulose substrates, and pretreatments are required before AD. Biological pretreatment is less expensive and more environmentally friendly than physical or chemical methods, but it has a longer cycle time and a more complex operation. The composite microbial community, which includes paddy soil microorganisms (Wang et al., 2021a), biogas slurry (Wang et al., 2020), and self-screened mixed strains (Yu et al., 2016), has demonstrated significant advantages in the hydrolysis and acidification (HA) process of lignocellulosic raw materials, which is based on the synergy of diverse microorganisms. The composite microbial community secretes extracellular enzymes to promote the hydrolysis of complex insoluble substrates (ISS) into soluble substances (SS), resulting in the production of volatile fatty acids (VFAs) (Hua et al., 2014), and the accumulation that could be used for methane production (especially in two-stage or three-stage AD reactor) or platform compound refinement (Fu et al., 2020). However, due to the simultaneous generation and consumption of intermediate products, the composite microbial community should be managed to hydrolyze ISS as much as possible to accumulate more SS and VFAs.

Recent research has gradually revealed benefits of aeration in HA for accelerating lignocellulose hydrolysis (Nguyen & Khanal, 2018). Moreover, the presence of oxygen can cause a portion of VFAs to be quickly oxidized rather than acetogenesis (Zhang & Jiang, 2019; Zhang et al., 2021). In general, anaerobic environment has an average ORP level of −300 mV or lower (Khanal & Huang, 2006; Krayzelova et al., 2015). However, some facultative anaerobic microbes benefit from increased VFAs accumulation when controlled in a microaerobic condition, which is characterized as 0.005–5 L O_2/L reactor/day (Nguyen & Khanal, 2018). In fact, the hydrolysis of lignocellulose requires more oxygen to promote the conversion of more ISS into SS (Qu et al., 2021; Zhang et al., 2019). Aerobic treatment in HA can be used as a pretreatment to accumulate more SS in a multi-stage AD reactor. There have been studies concentrating the interaction between aerobic conditions and the performance of HA of lignocellulosic materials to achieve the best treatment result. Existing studies mainly used micro-aeration for HA (Fu et al., 2020; Yu et al., 2016), whereas studies with more adequate aeration have shown a faster rate (approximately 24 h) of HA, as well as high ISS degradation (Luo et al., 2021; Zhang et al., 2019), but at the expense of SS consumption. However, there has been little research on aerobic HA of co-digestion systems with high solid content (total solid >15%), and it is unclear how combined aerobic and microaerobic treatment affects the balance of ISS degradation and SS consumption.

In this study, RS and PM were used as substrates, while biogas slurry was used as inoculum and water supplement to adjust the total solid because of the demand for waste recycling and utilization. A combination of aerobic and microaerobic treatment was used to investigate the degradation and consumption of ISS and SS during HA. Combined with the evolution of microorganisms, it is expected to provide fresh insights for HA in the biological treatment of mixed substrates with high solid content.

2. Material and methods

2.1. Rice straw, pig manure, and biogas slurry

RS was purchased from Lianyungang City, Jiangsu Province, China, and ground to 425 μm (60 meshed sieve). PM and biogas slurry was collected from a middle-scale pig farm (500 pigs) in Luxi county, Pingxiang City, Jiangxi Province, China, and stored in a dark polyethylene plastic bucket at 4 °C. PM was manually sieved to remove large particles of soil and hair, then ground to 425 μm (60 meshed sieve).

The physicochemical properties of RS and PM are shown in Table 1. The following are properties of biogas slurry: 7.04 in pH, 0.51% in total solid (TS), 89.79% in volatile solid (VS) (based on TS), 1020.52 mg/L in soluble chemical oxygen demand (sCOD), 161.40 mg/L in total organic carbon (TOC), 996.67 mg/L in total carbon (TC), 240.94 mg/L in total nitrogen (TN), 212.35 mg/L in ammonia nitrogen (NH_4^+–N).

2.2. Combination of aerobic and microaerobic treatment

The ratio of PM, RS, and biogas slurry was 1:1:6.5, with TS of approximately 15%. Experimental groups are shown in Table 2, and the entire HA lasted 14 days. Each replicate consisted of a 5-L aerobic reactor (1-L air/1000 g substrate/min) and nine 500-mL microaerobic reactors (stirred thrice per day) (Yu et al., 2017). The 5-L reactor was used to complete the aerobic treatment for 14 days and was recorded as the Ae group. From the first to the eighth day of aerobic treatment, a certain portion of the mixture was uniformly separated from the aerobic reactor and transferred to the other eight microaerobic reactors to complete second stage treatments, which were recorded as D1–D8 groups. The remaining microaerobic reactor was set up for 14 days of microaerobic treatment, which was recorded as the mAe group. Details of the aerobic and microaerobic treatment are provided in the supplementary material.

For each sample collection, 20 g of the mixture was collected from each beaker and centrifuged (8000 rpm, 15 min), and the separated solid and liquid were stored at −20 °C for further analysis.

2.3. Analytic methods

2.3.1. Solid fraction

Standard protocols for TS and VS were used for feedstocks and biogas slurry, following American Public Health Association methods 2540B and 2540E (A.D. Eaton et al., 2005). The elemental composition (carbon, nitrogen) was measured using an elemental analyzer (Elementar Vario EL cube, Germany). Cellulose, hemicellulose, lignin content, and particle size in substrates and HA residues were determined on the basis of a method using Vansoest et al. (1991) and laser particle size analyzer (BT-9300HT, Dandong Instrument Co., Ltd, Liaoning, China).

Table 1
Characteristics of rice straw (RS) and pig manure (PM) (Data are means ± standard deviation, n = 3).

Components	Rice straw	Pig manure
TS (%)	91.09 ± 0.43	36.01 ± 0.46
VS (% of TS)	83.42 ± 0.28	69.32 ± 0.58
C (% of TS)	43.77 ± 0.29	34.37 ± 0.50
N (% of TS)	0.86 ± 0.07	2.05 ± 0.12
Hemicellulose (% of TS)	36.65 ± 0.26	8.81 ± 0.03
Cellulose (% of TS)	24.77 ± 0.33	14.84 ± 0.15
Lignin (% of TS)	7.67 ± 0.23	2.82 ± 0.04

2.3.2. Liquid fraction

The pH and oxidation–reduction potential (ORP) were directly measured using a pH meter (PHS-3G, Shanghai INESA, China). sCOD and NH_4^+–N were determined using a water quality analyzer (5B–6C multi-parameter water quality analyzer, Lianhua Tech, China). TC, TN, and TOC were determined using a TOC/TN analyzer (Multi N/C® 3100, Analytik Jena AG, Germany). The reducing sugar was determined on the basis of the dinitrosalicylic acid (DNS) method (Lamounier et al., 2020). VFAs were measured based on the colorimetric ferric hydroxamate method with acetic acid as the standard (Chatterjee et al., 2018). CMCase and xylanase were determined on the basis of Hua et al. (2014) using the DNS method, and one enzyme activity unit (U) is defined as the amount of enzyme required to convert 1 μmol of a substrate within 1 min at 25 °C.

2.3.3. Microbial community structure

Based on results of sections 3.2, 0, 5, and 14 days in the D4 group were chosen for microbial diversity analysis. Each day, 10 mL of the mixture was separated and then stored in a −80 °C refrigerator for further analysis, which included fungi, bacteria, and archaea.

Total genomic DNA was extracted from the samples using an extraction kit (Shanghai Majorbio Bio-Pharm Technology, China) according to the manufacturer's instructions. The extracted DNA was checked using 1% agarose gel, and DNA concentration and purity were determined using NanoDrop 2000 UV-vis spectrophotometer (Thermo Scientific, Wilmington, USA).

Primers for amplifying the gene were as follows: The ITS1 region of Fungus ITF genes was amplified with ITS1F (5′-CTTGGTCATTTAGAGGAAGTAA-3′) and ITS2R (5′-GCTGCGTTCTTCATCGATGC-3′), and the V4 region of archaea and bacterial 16S rRNA was amplified with 515FmodF (5′-GTGYCAGCMGCCGCGGTAA-3′) and 806RmodR (5′-GGACTACNVGGGTWTCTAAT-3′). Amplification was completed by an ABI GeneAmp® 9700 PCR thermocycler (ABI, CA, USA) with the following parameters; 3 min at 95 °C, followed by 35 cycles (fungus) and 27 cycles (bacteria and archaea) of 30 s at 95 °C, 30 s at 55 °C, and 45 s at 72 °C. The Illumina Miseq sequencing service was provided by Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China). Finally, all sequences were clustered into operational taxonomic units (OTU) by setting a 3% distance limit.

2.3.4. Data analysis

The VS degradation in t day was calculated as Eq. (1).

$$\mathrm{VSdegradation}(\%) = \frac{(S_0 - S_t)}{S_0} \quad (1)$$

Where

S_0 represents the percentage of VS in 0 day (%, based on TS).

S_t represents the percentage of VS in t day (%, based on TS), $t \geq 1$.

The modified mass of hemicellulose, cellulose, and lignin was calculated based on the 100 g mixture Eq. (2).

$$\mathrm{Mass(g)} = C_{t,i} \times TS_t \times 100 \quad (2)$$

Where

$C_{t,i}$ represents the percentage of i in t day (%, based on TS), and i represents one of hemicellulose, cellulose, and lignin.

TS_t represents the TS in t day (%).

2.4. Statistical analysis

Three parallel replicates were performed in this study. Excel 2016 was used for data collation, and Origin 2019b and Adobe Illustrator 2020 were used for drawing. SPSS 23.0 software was used for one-way ANOVA to test the significance of the data and principal component analysis. Bioinformatics analysis was conducted on a platform (https://cloud.majorbio.com/) provided by Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China). The circus maps of microbial community structure were made online in http://mkweb.bcgsc.ca/tableviewer/visualize/, based on the method of Krzywinski et al. (2009).

3. Results and discussion

3.1. Variations in ISS and SS during HA

Fig. 1 showed indicators that represent the variation in ISS through VS degradation, and SS through sCOD, and other related indicators were shown in supplementary material. Overall, the VS degradation in the mAe group only achieved 18.24% in 14 days, whereas the Ae group reached 35.26%, indicating that the aerobic had a significant impact on ISS and SS and was also comparable with the previous results (Luo et al., 2021). Three periods could be divided during the HA based on ISS and SS trends.

3.1.1. First period: SS dissolution

In Fig. 1a, there was no change in sCOD over the first three days, implying that the total amount of SS had not been changed, i.e., the ISS had not been degraded, or that the generation and consumption rate of SS were flat during this stage. Furthermore, significant changes in VFAs (Fig. 1b), reducing sugar (Fig. 1c), and pH (see supplementary material) were observed in the first few days. VFAs significantly increased on the first day, from 7154 to 12515 mg/L in the mAe group and 10150 mg/L in the Ae group, whereas reducing sugar content decreased from 816 mg/L in the first three days. The VS degradation (Fig. 1d), was unnoticeable in the first two days, implying that the SS was primarily sourced from the dissolution of PM and RS during this period (0–2 days). Pre-degradation of RS and PM during storage accumulated some SS, such as reducing sugar, which was quickly leached and dissolved in the solution and converted to organic acids by numerous acidifying bacteria, resulting in a significant fall in the pH (Hua et al., 2014). Notably, 2–4 days were like a transition period, when ISS had begun to be degraded in the Ae group and sCOD of the D1 and D2 group were slightly higher than that of the Ae and mAe group, indicating the contribution of aerobic to ISS

Table 2
Experimental design.

Days Groups	1	2	3	4	5	6	7	8	9	10	11	12	13	14
mAe	*	*	*	*	*	*	*	*	*	*	*	*	*	*
D1	–	*	*	*	*	*	*	*	*	*	*	*	*	*
D2	–	–	*	*	*	*	*	*	*	*	*	*	*	*
D3	–	–	–	*	*	*	*	*	*	*	*	*	*	*
D4	–	–	–	–	*	*	*	*	*	*	*	*	*	*
D5	–	–	–	–	–	*	*	*	*	*	*	*	*	*
D6	–	–	–	–	–	–	*	*	*	*	*	*	*	*
D7	–	–	–	–	–	–	–	*	*	*	*	*	*	*
D8	–	–	–	–	–	–	–	–	*	*	*	*	*	*
Ae	+	+	+	+	+	+	+	+	+	+	+	+	+	+

Note: * represents the microaerobic treatment; + represents the aerobic treatment; – represents the aerobic treatment that was accomplished in the Ae group reactor.

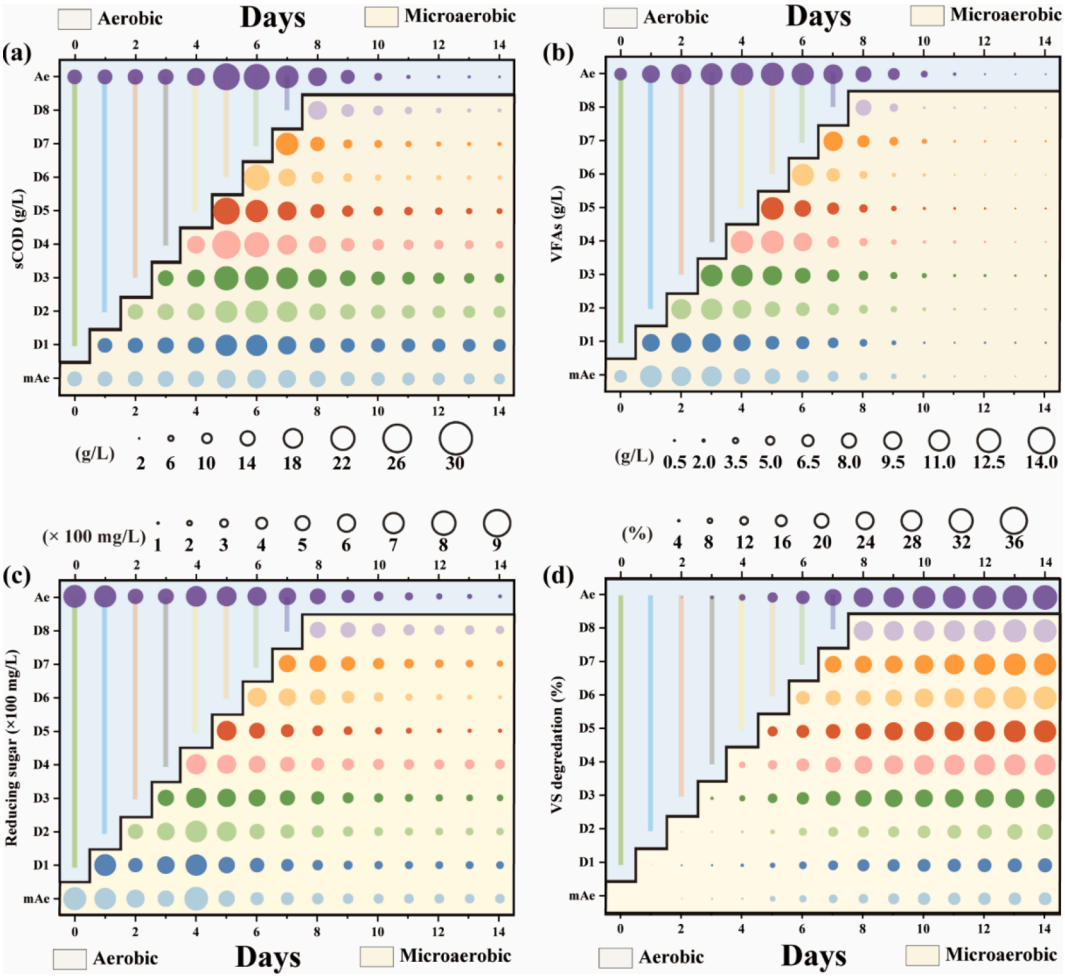


Fig. 1. Variations in (a) soluble chemical oxygen demand (sCOD), (b) volatile fatty acids (VFAs), (c) reducing sugar, and (d) volatile solid (VS) degradation during hydrolysis and acidification (HA) (Note: The upper left part represents the aerobic treatment, the lower right part represents the microaerobic treatment stage, and specific data were given in supplementary material).

degradation start-up and microaerobic to SS retaintion. Overall, the ISS degradation and SS consumption were at a low level during this period, whereas the main conversion was performed based on the preexisting leached SS.

3.1.2. *Second period: Degradation of ISS > consumption of SS*

The increased sCOD indicated that the hydrolysis rate of ISS in all groups was significantly greater than the SS consumption rate, entering the period of SS accumulation and all groups reached the peak on days 5–6. Furthermore, the mAe group had the lowest sCOD peak at 19226 mg/L on the sixth day, and the peak increased and advanced as the aerobic time increased. On day 5 of the D4 group, the maximum peak reached 29157 mg/L. The daily change of VS degradation, that is the VS degradation rate (see supplementary material) also reached a high level during the same period, resulting in the rapid conversion of ISS to SS. However, the degradation of lignocellulose resulted in the recovery of reducing sugar content, with a maximum content of 843 mg/L in the mAe group, which could be attributed to the more consumption of reducing sugar or conversion from reducing sugar to VFAs in other aerobic groups. Generally, VFAs are generated by reducing sugars via glycolysis in a lignocellulose environment (Nguyen & Khanal, 2018). On the second day, VFAs in the mAe group began to decline after reaching the peak, implying the inadequate production of VFAs precursor under microaerobic. However, VFAs in aerobically treated groups showed a plateau, with 11556–13228 mg/L during 2–6 days, which could be attributed to the higher SS content and preference of internal conversion from reducing sugar to VFAs under aerobic (Zhang et al., 2019). The ORP trough (see supplementary material) could also be considered microorganisms' strong activity to the degradation of ISS and the conversion of SS during this period (Yu et al., 2017).

The D4 group showed a pattern of sustained SS increase in the day after aeration stopped that differ from the groups D1-D3, implying the conducive to ISS degradation if the aerobic time could match the microbial growth and metabolism process. After entering the microaerobic stage with a slower SS consumption, the D4 group still retained a higher ISS degradation. Moreover, sCOD on fifth day in the D4 group surpassed that in the Ae group for the first time, implying that without continuous aeration, the sCOD and VFAs consumption rate of the D4 group may be lower than that of the Ae group. During this period, although aerobic resulted in a higher rate of ISS degradation and SS consumption, the gap between the two is greater than microaerobic, resulting in more accumulation of SS in the D4 group.

3.1.3. *Third period: Degradation of ISS < consumption of SS*

A significant decrease in sCOD, reducing sugars, and VFAs, were observed, indicating that the generation rate was lower than the consumption rate. The slower utilization of ISS by microorganisms implied that the consumption of the abovementioned SS became the primary metabolic process, resulting in alkali salt accumulation and pH recovery. Although microaerobic can ensure a low rate of SS consumption, all groups entered this period with the reduction of the easily degradable ISS and the proliferation of microorganisms. Simultaneously, a decrease

of NH_4^+–N (see supplementary material) was observed in the groups with longer aerobic treatment because aeration at higher pH could lead to the stripping of NH_4^+–N (Cao et al., 2019). Furthermore, aeration might have resulted in a similar A/O process along with the change of sCOD, which was not good for accumulating SS. Therefore, it was not suggested to continue the aeration or additional HA time considering the time and energy costs. Generally, during this period, the rate of SS generation was lower than the SS consumption mainly because of the decrease in biodegradable ISS, causing the content of SS to decrease in the entire system, which was no longer suitable for subsequent resourceization.

3.2. Evolution of different indicators with aerobic treatment time

In Fig. 2, mAe and D1–D4 can be grouped into one category based on the change in the loading plot, whereas D7, D8, and Ae can be grouped into another category. The image of groups D5 and D6 had significantly changed, resembling a transition, which could be due to the trend change of NH_4^+–N in these groups.

In addition to D6, most contributions to PC1 and PC2 originated from VS degradation and sCOD, respectively. As previously stated, VS degradation can be regarded as ISS, whereas sCOD can be regarded as SS. Therefore, SS contributed to PC1 and ISS degradation contributed to PC2 in mAe and D1–D4 groups, and ISS degradation contributed to PC1 and SS contributed to PC2 in the D5, D7, D8, and Ae groups. Furthermore, daily scores indicated that as the HA time increased, the degradation of ISS increased and SS first increased and subsequently reduced. The HA was mainly separated into three periods according to Section 3.1, of which the boundary between the second and third stages could be regarded as the balance point of ISS degradation and SS consumption, respectively. Cluster analysis was performed on the daily score of each group, illustrated as circles in Fig. 2. Days in blue circles were mainly concentrated in the areas with low ISS degradation and low SS content, whereas those in green circles were mainly concentrated in areas with high ISS degradation but low SS content. Therefore, the days in red circles could contain the balance point of degradation and consumption.

The main difference in red circles involved group D4 (on the 5th and 6th days), D5 (on the 5th day), and D6 (on the 5th and 6th days). The fifth and sixth days in the D4 group were under microaerobic conditions, whereas the fifth day in the D5 group, as well as the fifth and sixth day in the D6 group, were under aerobic conditions. Therefore, combined with Fig. 1, to obtain hydrolysate with the highest concentration of VFAs and sCOD with minimum aeration energy input, it is better to collect on the fifth day of the D4 group.

3.3. Degradation of ISS by microorganism

Throughout the HA, microorganisms degraded the lignocellulosic material, which could be considered the main ISS. The 0, 5, 9, and 14 days in groups Ae, mAe, and D4 were chosen to analyze the ISS degradation effect of a microorganism according to the results of Section 3.2.

As shown in Fig. 3a, only cellulose and hemicellulose were primarily degraded in all groups, whereas lignin was slightly destroyed in the Ae and D4 group. The degradation of lignin usually requires microorganisms, such as *Aspergillus* and *Trichoderma* (Wang et al., 2021b), but the microbial results in Section 3.4.2 did not reveal the presence of such bacteria in a large abundance. During the 0–5 days, the hemicellulose and cellulose could be degraded by 28.30% and 26.81% in the Ae group, respectively, and CMCase and xylanase in Fig. 3b also reached their peaks at 3.69 U/mL and 8.51 U/mL, respectively. During 5–14 days, when the rate of ISS degradation is less than SS consumption, the ninth day was chosen for further observation, and the degradation rate of ISS and enzyme activity began to decline as expected. Surprisingly, the activity difference between group D4 and Ae in CMCase was greater than that in xylanase on the ninth day, which was similar to the changes in the degradation rates of cellulose and hemicellulose. The degradation of cellulose and hemicellulose was indistinguishable under aerobic, but after transferring to microaerobic conditions, hemicellulose can be more biased utilized. Meanwhile, the change in D90 in the Ae group in supplementary material could show that aerobic can hydrolyze bigger particles. This could be because the aeration stones used in this treatment generate fine bubbles and promote continual particle movement in the system, allowing big particles to fully react with lignocellulosic breakdown enzymes.

3.4. Bioinformatics analysis

3.4.1. Microbial diversity

Mixtures from the D4 group at 0, 5, and 14 were chosen for fungus, bacteria, and archaea diversity analysis, respectively, because of the balance point on the fifth day in the D4 group.

As shown in Table 3, the obtained OTU could represent the genuine

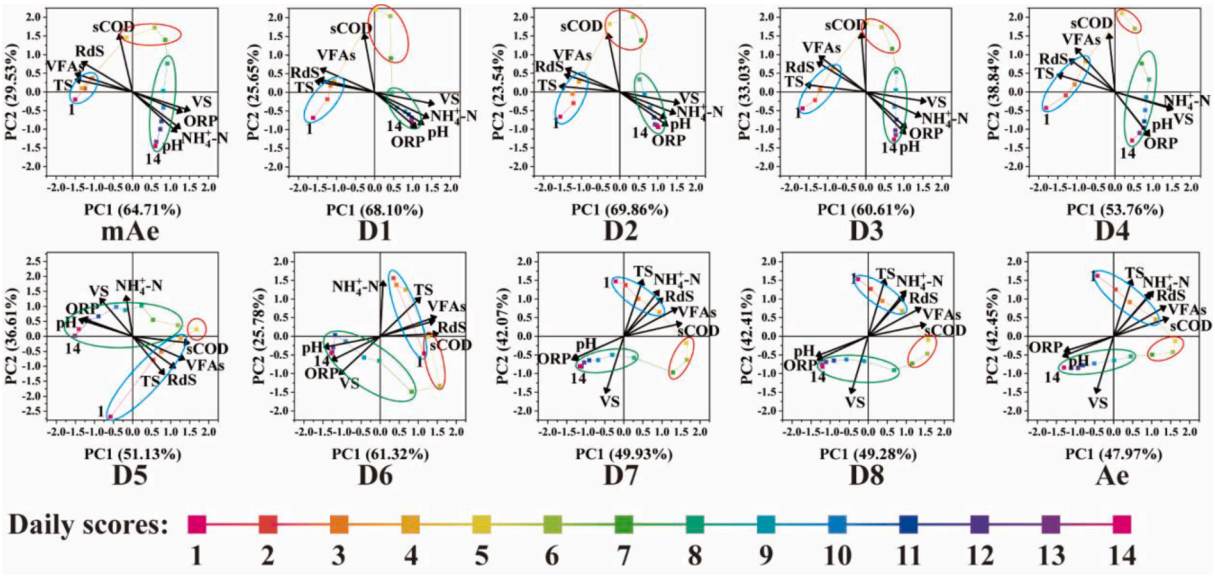


Fig. 2. Principal component analysis of indicators during 14 days of different treatments, and daily scores were shown as line-plots (Note: RdS means reducing sugar, volatile solid (VS) means VS degradation).

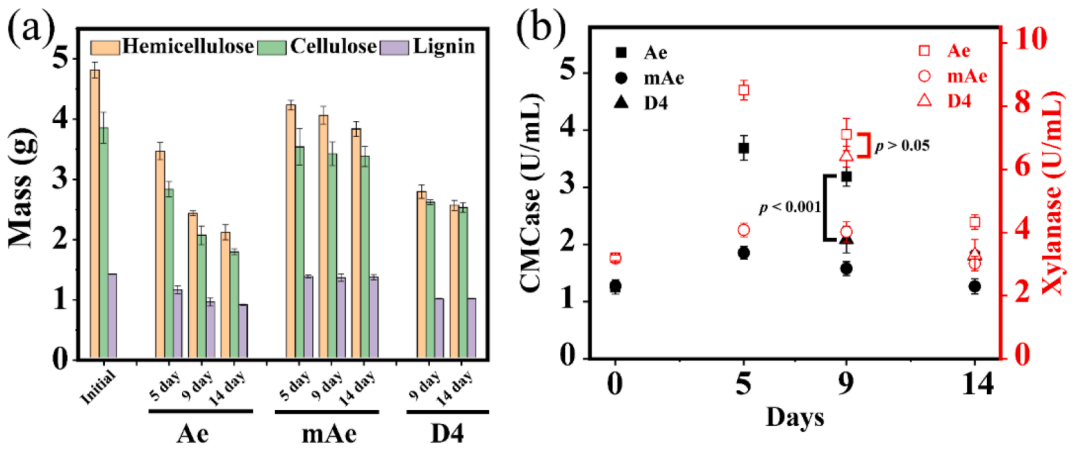


Fig. 3. (a) Lignocellulosic components and (b) CMCase and xylanase enzymes activity in 0, 5, 9, and 14 days in groups of Ae, mAe, and D4 (Data are presented as means ± standard deviation, n = 3).

Table 3
Alpha diversity of fungus, bacteria, and archaea of D4 group in 0, 5, and 14 days, respectively.

Samples		OTU	coverage	Community richness		Community diversity	
				Ace	Chao	Shannon	Simpson
Fungus	0 day	72,610	0.9999	96.2187	96.6732	2.1068	0.1734
	5 day	72,005	0.9999	54.5103	54.3333	1.0259	0.6100
	14 day	67,019	0.9998	45.7766	41.6752	1.3532	0.4354
Bacteria and Archaea	0 day	68,375	0.9995	600.1204	596.8814	4.0858	0.0528
	5 day	65,874	0.9984	466.9416	456.3448	3.2107	0.0718
	14 day	63,380	0.9988	427.9553	451.1818	3.3876	0.0735

samples (coverage >0.99). Fungus richness had been declining, whereas the diversity first decreased and then increased. The richness of fungus was initially high because of numerous sources of materials. Because most fungus are aerobic microorganisms in this study (Li et al., 2016), the flooded state would hinder fungus growth. The richness reduced and the diversity grew after the balance point, which could be explained by some fungus with a higher abundance declining more because of the lack of oxygen, and the abundance of some fungus with increasing lower abundance.

In the first five days, the richness and diversity of bacteria and archaea significantly decreased. This could be attributed to two factors. First, aerobic therapy was extremely harmful to most archaea (Nguyen & Khanal, 2018). Second, the effect of aeration treatment promotes the enrichment of certain bacteria, which played an important role in the HA process (Wang et al., 2021a). The richness and diversity did not vary significantly between 5 and 14 days. After aeration was turned off, the entire solution was full of SS, including reducing sugar and VFAs that bacteria and archaea could use.

3.4.2. Microbial community structure

In Fig. 4a, top seven fungus genera were *Candida*, *Trichosporon*, *Pseudallescheria*, *Scedosporium*, *Eleutherascus*, f_*Dipodascaceae,* and *Geotrichum* (accounted for 97.83% of the total OTU), the abundance of which significantly changed in three days. On day 0, *Pseudallescheria* (44.43%), *Scedosporium* (32.94%), and *Eleutherascus* (12.29%) were the main types of fungus that mainly existed in the original substrates and had been proven to degrade lignocellulose, organic matter, and hydrocarbons (Chen et al., 2013; Kraková et al., 2016; Ros et al., 2010). However, the abundance of these three fungus fell greatly during 0–5 days, showing that they cannot adapt to the liquid environment under aeration or acidic solution; however, they may be effective in the degradation of ISS in PM and RS to SS during storage. *Candida* had the highest abundance (77.57%) at 5 days. Previous research has shown that the species *Candida tropicalis* is a straw-decomposing fungus that can use hemicellulose to generate ethanol and xylitol (Swain & Krishnan, 2015; Zhuang et al., 2009), and it was further proved to be useful in the aerated liquid condition in this study. *Trichosporon* (64.37%), f_*Dipodascaceae* (15.91%), and *Geotrichum* (4.67%) were the main fungus in 14 days. Among them, *Trichosporon* and *Geotrichum* are proved as yeast (Huang et al., 2014; Yan et al., 2019), and f_*Dipodascaceae* was discovered in brewing and metabolism of reducing sugar (Cao et al., 2021; Li et al., 2016), which indicates they can metabolize the SS in a microaerobic environment.

Community diversity at the genus level was more prevalent in bacteria and archaea than in fungus because of high tolerance. The top 12 genera account for 70.26% of all OTUs, with 10 bacteria and 2 archaea. The bacteria *Clostridium_sensu_stricto_1* (8.68%), *Advenella* (6.10%), *Enterococcus* (5.67%), *Fermentimonas* (5.67%), and archaea *Alkaliflexus* (18.29%) and *Escherichia-Shigella* (5.71%) were the main genus in origin substrates, which could stably coexist in the original substrate and had been proven to secrete an enzyme to utilize lignocellulose (Coenye et al., 2005; Gao et al., 2014), convert carbohydrates and other SS to VFAs, H_2, and CO_2 (Hahnke et al., 2016; Liu et al., 2022; Valdez-Vazquez et al., 2017). On 5 days, the main genus was *Lactobacillus* (24.26%), *Clostridium_sensu_stricto_1* (17.52%), *Acetobacter* (13.90%), and *Bifidobacterium* (11.85%), all of which play important roles in the conversion of SS to VFAs in the aerated liquid environment (Moniz et al., 2016; Zhao et al., 2018). After the balance point, reduction of precursor substances of VFAs resulted in a significant decrease in the abundance of the above acid production genus. Finally, at 14 days, bacteria such as *Clostridium_sensu_stricto_1* (22.08%) and *Prevotella* (15.97%) (Jose et al., 2017) that can degrade lignocellulose under very low oxygen were the main group, and there was also a part of the genus, such as *Acetobacter* (8.55%), *Terrisporobacter* (6.87%) and *Turicibacter* (5.71%) that can continue to produce acetic acid, propionic acid, and butyric acid (Guan et al., 2021; Pore et al., 2019) under same conditions.

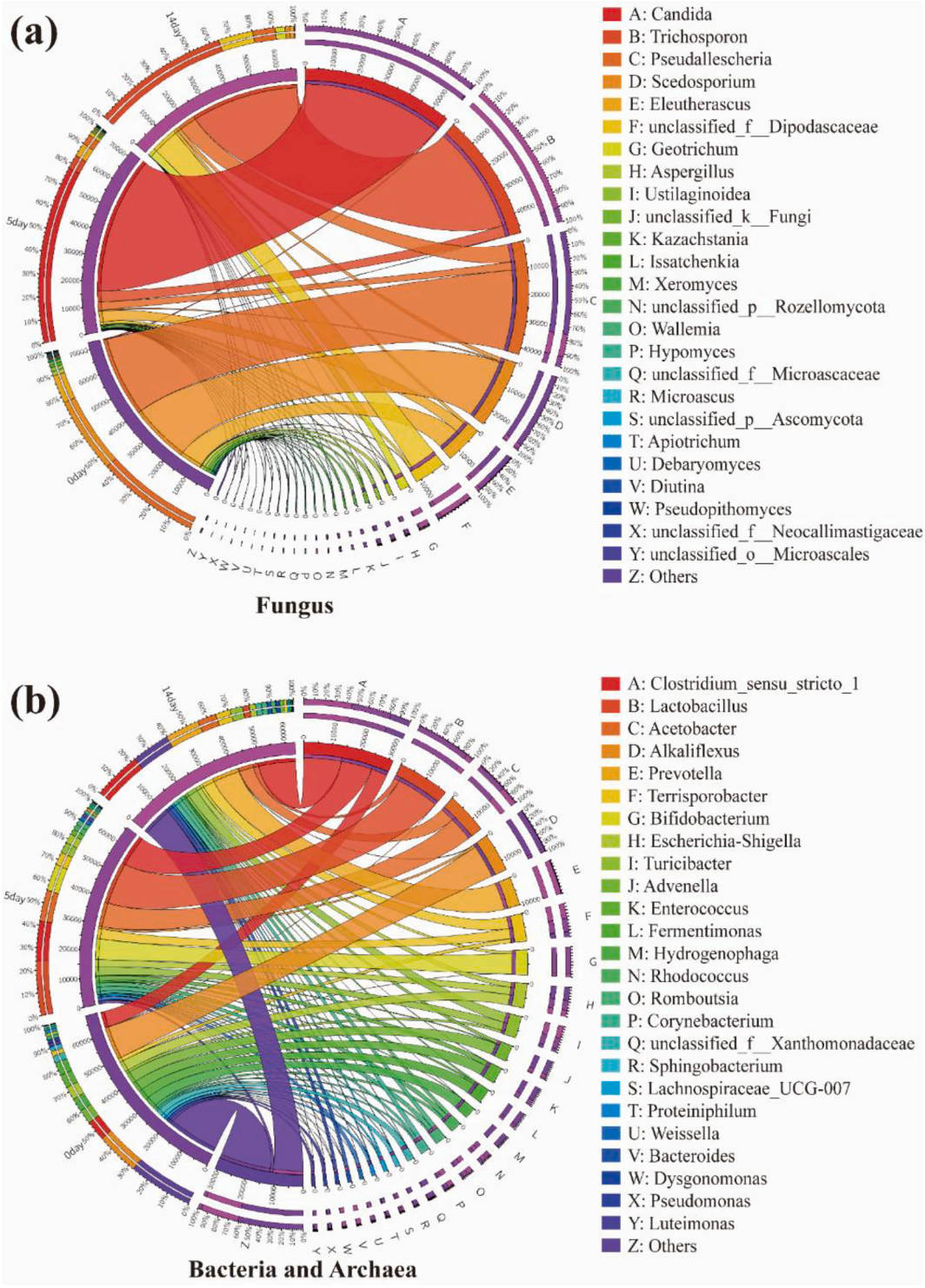


Fig. 4. Microbial community diversity of (a) fungus, (b) bacteria, and archaea in 0, 5, and 14 days of Ae, mAe, and D4 groups at the genus level [The top 25 based on total genus operational taxonomic units (OTU)].

3.4.3. Microbial community structure and substrate conversion

Fig. 5a shows that the common OTU accounted for 97.83%, indicating that the combination of aerobic and microaerobic did not result in significant changes in fungal types. However, the common portion accounted for 83.84% of bacteria and archaea in Fig. 5b, implying the significant influence of oxygen, especially the common part of 5 and 14 days (8.05%), and a unique part of 14 days (6.15%).

The significantly increased fungus, during 0–5 days, were *Candida, Trichosporonhe, Dipodascaceae,* and *Geotrichum*. Moreover, the bacteria and archaea were *Lactobacillus, Bifidobacterium, Clostridium_sensu_stricto_1, Terrisporobacter, Turicibacter, and Acetobacter*, proving that these genera might be effective for the degradation of ISS and generation of SS. Furthermore, the abundance of some of the abovementioned genera, such as *Trichosporonhe, Dipodascaceae, Geotrichum, Clostridium_sensu_stricto_1, Terrisporobacter, Turicibacter,* and *Prevotella*, increased within 5–14 days. This was attributed to the high tolerance and capacity of these genera to grow in microaerobic conditions. However, the reduction of SS might be attributed to these genera, which took excessive

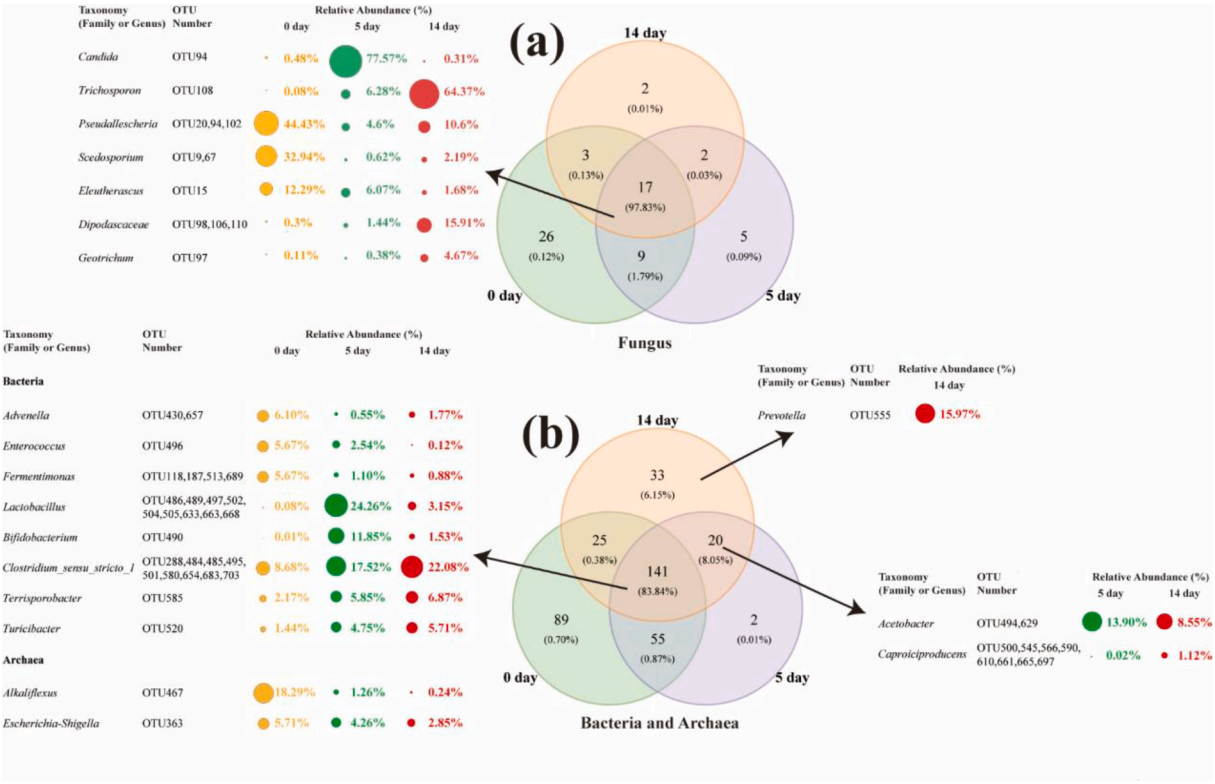


Fig. 5. Venn diagram of (a) fungus, (b) bacteria, and archaea based on operational taxonomic units (OTU) in 0, 5, and 14 days, and the genus distributions of the unique and common-shared part. Only OTU with relative abundance >1% at the genus level are included, other parts with total OTU >1% are presented in supplementary material (The number in the Venn diagram represents the genus number of this part, and the parenthesis represents the percentage of the total OTU that accounts for the genus).

consumption of organic matter to complete their growth. However, the genus, which included *Candida, Lactobacillus, Bifidobacterium,* and *Acetobacter*, declined during 5–14 days, which could explain their positive contribution to the degradation of ISS and generation of VFAs. The high abundance values of these four genera can be maintained in the HA reactor, which may be more favorable to the accumulation of highly SS and VFAs.

3.5. Significance

The mixed system of PM and RS mainly encountered three distinct periods during 14 days of HA, such as SS dissolution, ISS degradation > SS consumption, and ISS degradation < SS consumption. The best liquid performance was attained on the fifth day of the D4 group, which also served as a balance point. Aerobic showed an excellent ability to degrade ISS over the first 4 days, followed by one day of microaerobic treatment that maintained a favorable SS accumulation state, avoiding the excess consumption of SS induced by continued treatment.

Although aerobic accelerates ISS degradation and SS consumption, more emphasis should be devoted to the balance between degradation and consumption in future studies, which could help achieve accurate control of SS accumulation. Simultaneously, HA evaluation based on a certain indicator should be avoided, similar to VFAs of the mAe group in this study, which attained the maximum value on the first day but did not reach the peak of the ISS degradation rate, resulting in misjudgment. Compared to previous studies (Table 4), the treatment protocol in this study achieved very good performance of sCOD and VFAs at a 15% TS. In addition, this study proposed a new biogas slurry recycling scheme rather than be treated as waste (Gu et al., 2021; Zhang et al., 2020). Re-eutrophication of biogas slurry is not only achieved for biogas production (Yu et al., 2016) or platform chemical production (Alonso et al., 2015) but also ammonia is partially removed by aeration.

Table 4
Comparison of aerobic applications for hydrolysis and acidification (HA) of lignocellulose.

Waste	TS	Aerobic conditions	Results
Oilseed rape straw[1]	6%	20 mL O_2/g VS/d 10 d	VFAs: 4667.3 ± 76.7 mg/L sCOD: 6792.6 ± 167.2 mg/L
Corn stover[2]	8%	2.5 L air/min/980 $g_{substrate}$ 24 h	C2–C5 acids: ≈350 mmol/L
Corn stover pith[3]	8%	0.5 mL air/L 16 h	C2–C5 acids: 165 mmol/L Soluble sugar: 25 mg/L
Rice straw[4]	8%	0.3 L air/min/L 24 h	C2–C5 acids: 6225.15 mg/L $VS_{removal}$: 50.49%
Corn straw[5]	4.44%	5 mL O_2/g VS 5 d	VFAs: ≈2500 mg/L
Corn stalk[6]	10%	Open without seal 9 d with intermit stirring	C1–C4 and Lactic acids: ≈13000 mg/L
Rice straw and pig manure[7]	15%	1 L air/1000 $g_{substrate}$/min for 4 day and open in the air for 1 day	sCOD: 29157 mg/L VFAs: 13219 mg/L

Note: Datas were obtained from below study. 1, (Wang et al., 2020); 2, (Qu et al., 2021); 3, (Zhang et al., 2019); 4, (Luo et al., 2021); 5, (Fu et al., 2020); 6, (Yu et al., 2016); 7, This study.

4. Conclusions

Combined aerobic and microaerobic treatment during the 14 days resulted in the HA of PM and RS. Aerobic improved the ability of microorganisms to degrade the substrate and hydrolyze big particle size lignocellulose, with *Candida*, *Lactobacillus*, *Bifidobacterium*, and

Acetobacter playing major roles in degrading insoluble substrates and accumulation of VFAs. sCOD, VFAs, reducing sugars, and VS degradation rate all attained peaks within 4–6 days, in which insoluble substrate degradation and soluble substrate consumption were in balance. Furthermore, sCOD (29157 mg/L) and VFAs (13219 mg/L) were obtained for 4 days aerobic, following microaerobic for 1 day.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2022.126880.

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